

Research for Back tester Project: FIFO operations in Memory

1) What is the First in First Out Event Queue

This event queue is vital to prevent look ahead bias. This is when an entire CSV file is entered into a back testing system. This means that when the code is looking into data, it can 'peek' at data that would not have been available at that time. This is the essence of look ahead bias.

Using a FIFO event queue, the backtester is mathematically forced to process market data chronologically, tick by tick. A tick is the smallest possible unit of measurement in the financial markets. It represents a single update to an order book.

2) Relevancy to code:

In terms of the code for the backtester, this means the data handler must read the Monday price, push it to the FIFO event queue. The system then calculates the math, checks its portfolio balance, and then decides when to buy or sell. This is only based on Monday's data. It then can loop forward to read Tuesday's row.